# Project Proposal Template

## Project:

Smart Medical Diagnostics & Risk Scoring System

**Objective:**.

Professional Data Analytics (DEPI – Round 5)

**Description:**

The goal of this project is to develop an end-to-end data analytics and predictive system that automates emergency department patient triage and risk classification using clinical data (MIMIC-IV Clinical Demo Dataset). The system addresses operational delays in identifying high-risk patients, provides dynamic risk scoring based on real-time vital signs and patient history, and optimizes clinical throughput and resource allocation across emergency care units.

**Tools & Design:**

Python Data Pipeline: Built with OOP, custom exception handling, and automated activity logging.

Data Warehouse**:** Relational SQL schema for structured clinical data storage and analytical queries.

Power BI Suite: Multi-page interactive dashboard covering Triage, Patient, and Executive views.

**Technologies to Use:**

Python , SQL Server , Power BI , Jupyter Notebook

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**Team Members & Responsibilities:**

1- Mennatullah Abdul Monem Samy Abdul Monem

2- Shahd Ashraf Hassan Abdelrahman

3- Fady Talaat Edwar Henien

4- Hossam Ramzi mohammed elbohoty

5- Menna Mohamed ali

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## Week 1: *[Tasks Description With Team Member Responsibility]*

- **[**Shahd Ashraf**]:** Import raw MIMIC-IV clinical triage and vital sign datasets into SQL Server.
- **[**Menna Mohamed**]:** Perform preliminary data cleaning and database initialization in SQL Server.
- **[**Mennatullah Abdul Monem**]:** Design the relational database schema (ERD) and set up primary/foreign key relationships.
- **[**Hossam Ramzi**]:** Write and execute SQL queries, joins, and CREATE VIEW statements for unified patient data.
- **[**Fady Talaat**]:** Define project scope, requirements, and set up the shared GitHub repository structure.

**Deliverables:**

- Fully structured SQL database schema and tables in SQL Server.
- Unified SQL patient views for integrated clinical analytics.
- Initial database ERD and architecture design document.

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## Week 2: *[Tasks Description With Team Member Responsibility]*

- **[**Mennatullah Abdul Monem**]:** Extract the unified patient dataset from SQL Server into Python for advanced processing.
- **[**Fady Talaat**]:** Perform comprehensive **Feature Engineering** to derive vital sign metrics, age brackets, and clinical risk indicators.
- **[**Menna Mohamed**]:** Implement Python OOP classes and exception handling for advanced data validation.
- **[**Shahd Ashraf**]:** Initialize, train, and evaluate the machine learning risk classification model in Python.
- **[**Hossam Ramzi**]:** Validate model accuracy and assign emergency triage acuity risk scores (High, Moderate, Low Risk).

**Deliverables:**

- Feature-engineered clinical dataset ready for predictive analytics.
- Modular Python feature engineering and machine learning pipeline scripts.
- Trained ML risk classification model with performance evaluation metrics.

## Week 3: *[Tasks Description With Team Member Responsibility]*

- **[**Mennatullah Abdul Monem**]:** Connect Power BI directly to the SQL Server Data Warehouse and model output tables.
- **[**Fady Talaat**]:** Build View 1 (Triage View) in Power BI for real-time emergency priority queues and patient risk alerts.
- **[**Menna Mohamed**]:** Build View 2 (Patient View) to track individual longitudinal vital sign trends over time.
- **[**Shahd Ashraf**]:** Design data relationships, custom DAX measures, and clinical KPIs in Power BI.

**Deliverables:**

- Connected Power BI project workspace file linked to SQL Server.
- Fully operational Triage View dashboard for emergency priority queueing.
- Interactive Patient View dashboard for historical vital sign tracking.

## Week 4: *[Tasks Description With Team Member Responsibility]*

- **[**Mennatullah Abdul Monem**]:** Build View 3 (Executive View) for hospital bed occupancy, throughput, and operational metrics.
- **[**Fady Talaat**]:** Conduct end-to-end system testing and pipeline verification between SQL, Python, and Power BI.
- **[**Menna Mohamed**]:** Perform code refactoring, script cleanup, and GitHub repository organization.
- **[**Shahd Ashraf**]:** Write the comprehensive final project report, user guide, and technical documentation.
- **[**Hossam Ramzi**]:** Design and format the final PowerPoint presentation deck for project defense.

**Deliverables:**

- Complete multi-page Power BI clinical analytics dashboard suite (Triage, Patient, Executive).
- Fully integrated, tested, and documented End-to-End Analytics System.
- Final project report document and PowerPoint presentation slides.